Two-row recurrence for Koornwinder polynomials

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Koornwinder polynomials

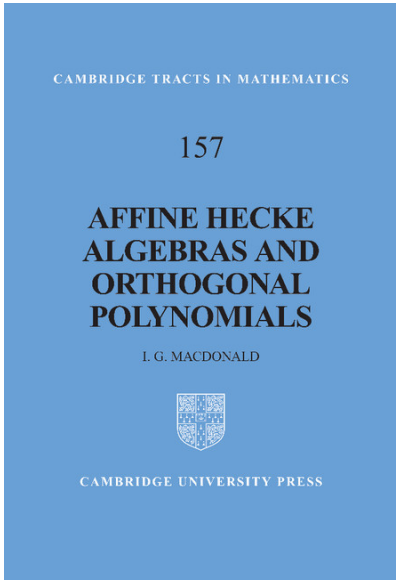
Koornwinder polynomials are Laurent polynomials in n variables, depending on six parameters q, t, a, b, c, d.

- Koornwinder '92: symmetric polynomials, indexed by partitions, introduced as multivariable orthogonal polynomials generalizing the Askey–Wilson polynomials (one variable).
- Sahi '99, '00: double affine Hecke algebra of type $(C^\vee, C)$, used to prove Macdonald's duality conjecture for Koornwinder polynomials and to define the nonsymmetric Koornwinder polynomials.

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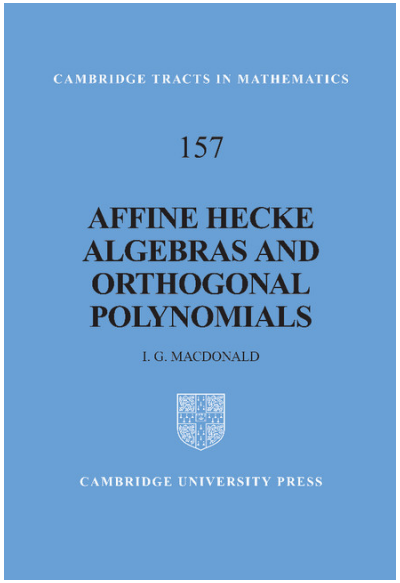


Macdonald '03

Koornwinder polynomials

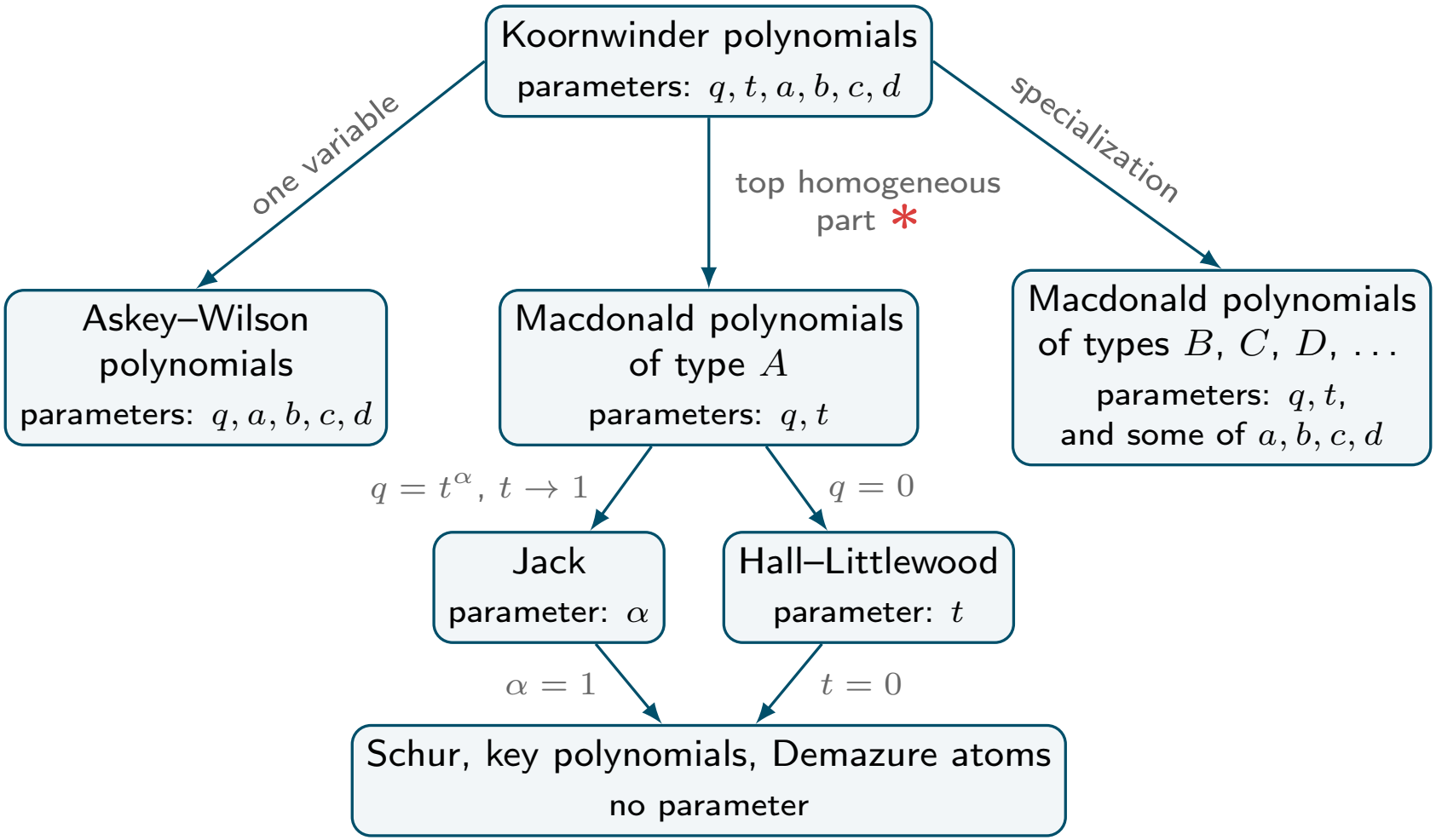
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- Cantini–Garbali–de Gier–Wheeler '16: connected Koornwinder polynomials to interacting particle systems (multispecies ASEP with open boundaries).



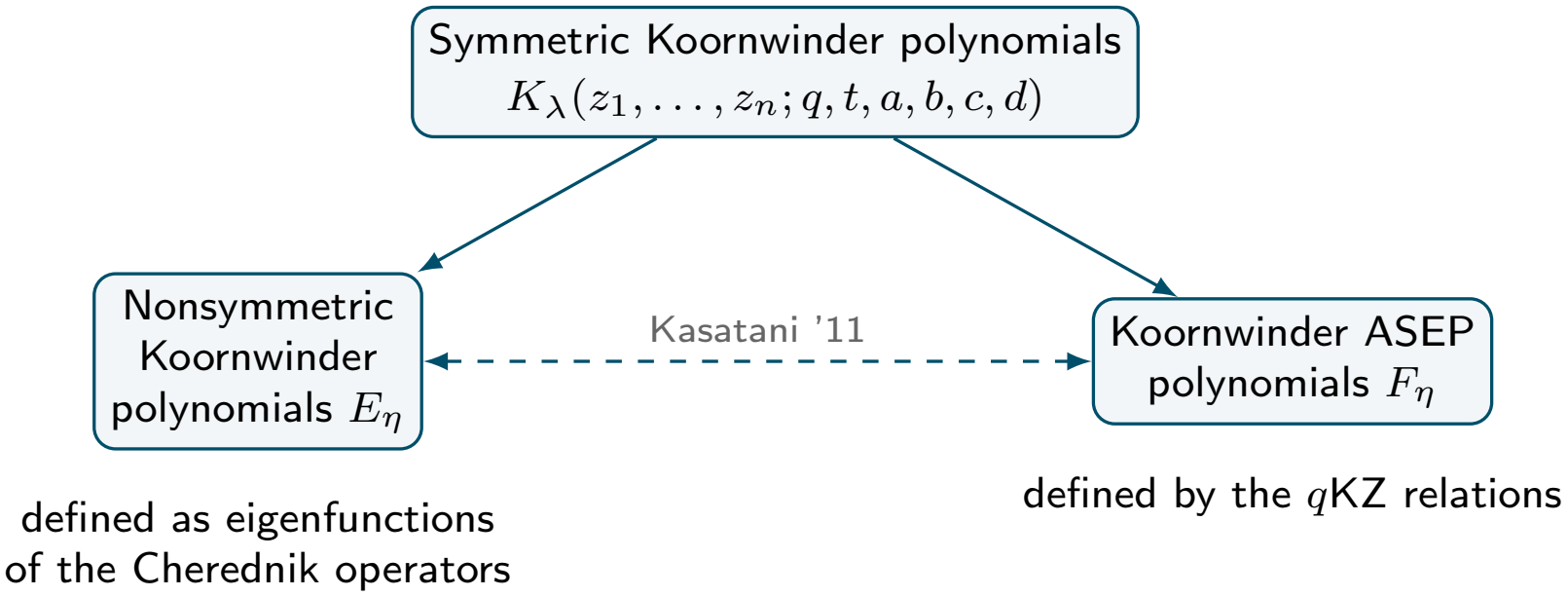
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A family of families

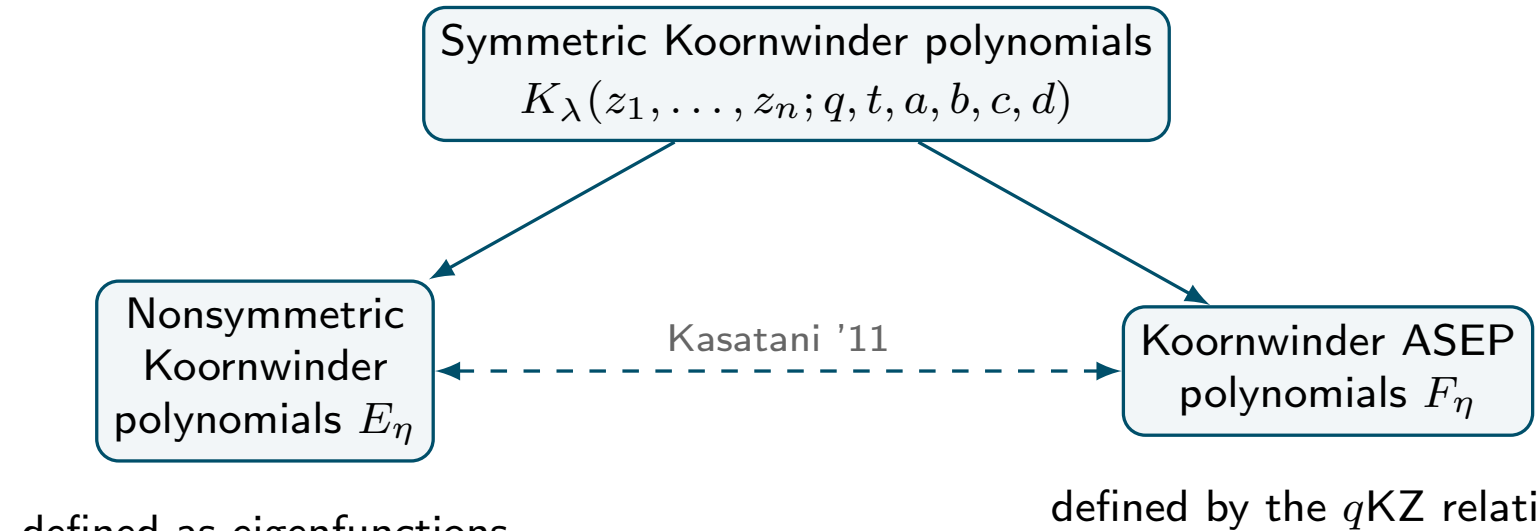


* Symmetric case: van Diejen '95. Nonsymmetric case: BD–Williams.

Two nonsymmetric refinements of the symmetric Koornwinder polynomials

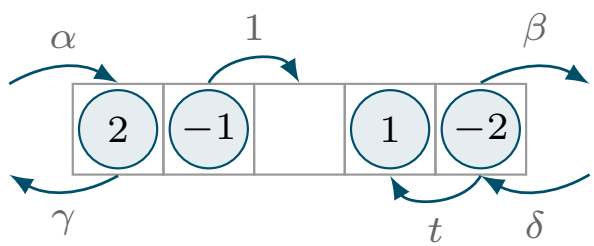


Two nonsymmetric refinements of the symmetric Koornwinder polynomials



Theorem (Cantini–Garbali–de Gier–Wheeler '16)

For suitable a, b, c, d depending on the boundary rates, the stationary probability of a state μ of the multispecies ASEP with open boundaries on n sites is proportional to $F_\mu(z_1 = \cdots = z_n = 1; q = 1, t, a, b, c, d)$.



Some examples of Koornwinder ASEP polynomials for $n = 3$

$$F_{(0,1,0)} = z_2 + \frac{t\big(t^2 ac(b+d) - a - c\big)}{1 - t^4 abcd}$$

Some examples of Koornwinder ASEP polynomials for $n = 3$

$$F_{(0,1,0)} = z_2 + \frac{t\big(t^2ac(b+d) - a - c\big)}{1 - t^4abcd}$$

$$\begin{aligned} F_{(0,1,-1)} = {} & z_2 z_3^{-1} + \frac{tbd(a+c) - b - d}{1 - t^2abcd}\, z_2 + \frac{t\big(tac(b+d) - a - c\big)}{1 - t^2abcd}\, z_3^{-1} \\ & + \frac{t}{(1 - t^2abcd)(1 - t^3abcd)} \Big((a+c)(b+d) - t\big(ac(b+d)^2 + bd(a+c)^2 - abcd\big) \\ & \quad + t^2abcd\big((a+c)(b+d) - 1\big) - t^3(1-t)(abcd)^2 \Big) \end{aligned}$$

In the cases of nonnegative compositions, the top degree part is represented in red. We will see that it corresponds to type A ASEP polynomials.

Some examples of Koornwinder ASEP polynomials for $n = 3$

$$
\begin{aligned}
F_{(0,2,0)} = {} & z_2^2 + \frac{1-t}{1-qt}\left(z_1 + qz_3\right)z_2 + \frac{q(1-t)}{1-qt}\left(z_1^{-1} + z_3^{-1}\right)z_2 + \frac{qt\left(qt^2ac(b+d) - a - c\right)}{1-q^2t^4abcd}z_2^{-1} \\
& + \frac{qt(1-t)\left(qt^2ac(b+d) - a - c\right)}{(1-qt)(1-q^2t^4abcd)}\left(z_1 + q\,z_1^{-1} + z_3 + z_3^{-1}\right) \\
& - \frac{1}{(1-qt)(1-q^2t^4abcd)}\Big(q^3t^5\left(ac(b+d) + bd(a+c)\right) + q^2t^4ac(b+d) - q^2t^3ac(b+d) \\
& \quad - q^2t^2bd(a+c) - q^2t^3(a+b+c+d) - qt^2ac(b+d) - qt^2(a+c) \\
& \quad + qt(a+c) + q(b+d) + a + c\Big)z_2 \\
& - \frac{1}{(1-qt)(1-qt^4abcd)(1-q^2t^4abcd)}\Big(q^4t^6abcd\big(t^3abcd + t^2(a+c)(b+d) \\
& \quad - t^2abcd + t^2ac + tabcd - 1\big) \\
& \quad + q^3t^4\big(t^4a^2c^2(b+d)^2 - t^4a^2bc^2d - t^3a^2b^2c^2d^2 - t^2bd(a+c)^2 - t^2ac(a+c)(b+d) \\
& \quad - t^2ac(b+d)^2 - tabcd(a+c)(b+d) + t^2abcd - ta^2bc^2d - tabcd + abcd\big) \\
& \quad - q^2t^2\big(t^4ac(a+c)(b+d) + t^3a^2c^2(b+d)^2 - t^3a^2bc^2d + t^3abcd \\
& \quad - t^2(a+c)^2 - t^2(a+c)(b+d) - t^2abcd - tbd(a+c)^2 - tac(a+c)(b+d) \\
& \quad - tac(b+d)^2 + t^2ac + tabcd - 1\big) \\
& \quad + q\big(t^3ac(a+c)(b+d) + t^4ac + t^3abcd - t(a+c)^2 - t(a+c)(b+d) \\
& \quad - t^2 + tac + t - 1\big) - tac\Big)
\end{aligned}
$$

Some examples of Koornwinder ASEP polynomials for $n = 3$

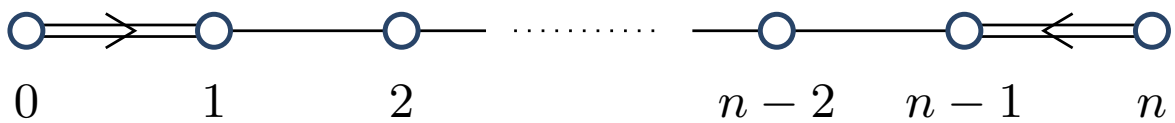
$$F_{(0,2,1)} =$$

$$\begin{aligned}
&z_2^2 z_3 + \tfrac{(1-t)}{(1-qt^2)} z_1 z_2 z_3 + \tfrac{1}{(1-qt^2)(1-t^2abcd)(1-q^2t^4abcd)} \big(q^3t^8a^2b^2c^2d + q^3t^8a^2b^2c^2d^2 + q^3t^7a^2b^2cd^2 + q^3t^7ab^2c^2d^2 + q^2t^7a^2b^2c^2d + q^2t^7a^2bc^2d^2 - q^3t^7a^2bcd - q^3t^7abc^2d - q^2t^6a^2b^2c^2d - q^2t^6a^2bc^2d^2 + q^3t^6a^2bcd + \\
&q^3t^6abc^2d - q^2t^6a^2bcd - q^2t^6abc^2d - q^2t^4a^2b^2cd^2 - q^2t^4ab^2c^2d^2 + q^2t^5a^2bcd - q^2t^5ab^2cd + q^2t^5abc^2d - qt^4a^2b^2c^2d - q^2t^5abcd^2 - qt^4a^2bc^2d^2 - q^3t^5abc - q^3t^5abd - q^3t^5acd - q^3t^5bcd - q^2t^4a^2bcd - qt^5a^2bcd + \\
&q^2t^4abc^2d - qt^5abc^2d + qt^4a^2bcd + qt^4abc^2d - q^2t^4abc - q^2t^4acd + q^2t^3abc + q^2t^3acd + qt^2ab^2cd + qt^2abcd^2 - qt^3abc + q^2t^2abd - qt^3acd + q^2t^2bcd + t^2a^2bcd + t^2abc^2d + q^2t^3a + q^2t^3b + q^2t^3c + 2qt^2abc + \\
&q^2t^3d + 2qt^2acd + qt^3a + qt^3c - qta - qtc - qb - qd - a - c\big) z_2 z_3 + \tfrac{t(tabc+tacd-a-c)}{(1-t^2abcd)} z_2^2 + \tfrac{(1-t)qt^2(qt^2abc+qt^2acd-a-c)}{(1-qt^2)(1-q^2t^4abcd)} z_1 z_3 - \tfrac{t(1-t)}{(1-qt^2)(1-t^2abcd)(1-q^2t^4abcd)} \big(q^2t^5a^2b^2c^2d + q^2t^5a^2bc^2d^2 - q^2t^5a^2bcd - q^2t^5abc^2d + \\
&q^2t^4abc + q^2t^4acd - q^2t^3abc - q^2t^3acd + qt^3abc + qt^3acd - qt^2abc - qt^2acd - qt^2a - qt^2c - tabc - tacd + qta + qtc + a + c\big) z_1 z_2 + \tfrac{(1-t)q}{(1-qt^2)} z_1^{-1} z_2 z_3 + \tfrac{1}{(1-qt^2)(1-t^2abcd)(1-qt^3abcd)(1-q^2t^4abcd)} \big(q^4t^{10}a^3b^3c^3d^3 + \\
&q^4t^{10}a^3b^3c^2d^2 + q^4t^{10}a^3b^2c^3d^2 + q^4t^{10}a^2b^3c^3d^2 + q^4t^{10}a^3b^2c^2d^3 + q^4t^{10}a^2b^2c^3d^3 + q^3t^{10}a^3b^3c^3d + q^3t^{10}a^3b^2c^3d^2 + q^3t^{10}a^3b^3c^2d^2 + q^3t^9a^3b^2c^3d^2 - q^4t^9a^2b^2c^2d^2 - 2q^3t^8a^3b^2c^3d^2 - \\
&q^3t^9a^3b^2c^2d - q^3t^9a^2b^2c^3d - q^3t^9a^3bc^2d^2 - q^4t^8a^2b^2c^2d^2 - q^3t^7a^3b^3c^2d^2 - q^3t^9a^2bc^3d^2 - q^3t^7a^2b^3c^3d^2 - q^3t^7a^3b^2c^2d^3 - q^3t^7a^2b^2c^3d^3 - q^3t^8a^2b^3c^2d - q^2t^7a^3b^3c^3d - q^3t^8a^3b^2c^2d - 3q^3t^8a^2b^2c^2d^2 - \\
&q^3t^8ab^2c^3d^2 - q^2t^7a^3b^2c^3d^2 - q^3t^8a^2bc^2d^3 - q^2t^7a^3bc^3d^3 - q^4t^8a^2b^2c^2d - q^4t^8a^2b^2c^2d^2 - q^4t^8ab^2c^2d - q^2t^8a^3b^2c^2d - q^2t^8a^2b^2c^3d - q^4t^8a^2bcd^2 - q^4t^8abc^2d^2 - q^2t^8a^3bc^2d^2 - q^3t^7a^2b^2c^2d^2 - q^2t^8a^2bc^3d^2 + \\
&q^3t^8a^2bc^2d + 2q^3t^6a^2b^2c^2d^2 - q^3t^7a^2b^2c^2 - 3q^3t^7a^2bc^2d + q^2t^6a^3b^2c^2d + q^2t^6a^2b^2c^3d - q^3t^7a^2c^2d^2 + q^2t^6a^3bc^2d^2 - q^2t^6a^2b^2c^2d^2 + q^2t^6a^2bc^3d^2 + q^4t^7abcd + q^2t^7a^3bcd + 2q^3t^6a^2bc^2d + 2q^2t^7a^2bc^2d + \\
&q^2t^5a^2b^3c^2d + q^2t^7abc^3d + q^2t^5a^3b^2cd^2 + 4q^2t^5a^2b^2c^2d^2 + q^2t^5ab^2c^3d^2 + q^2t^5a^2bc^3d^2 + q^2t^6a^2b^2c^2 + q^3t^5a^2b^2cd + q^2t^6a^2b^2cd - 3q^2t^6a^2bc^2d + q^3t^5ab^2c^2d + q^2t^6ab^2c^2d + qt^5a^3b^2c^2d + qt^5a^2b^2c^3d + \\
&q^3t^5a^2bcd^2 + q^2t^6a^2bcd^2 - q^2t^6a^2c^2d^2 + q^3t^5abc^2d^2 + q^2t^6abc^2d^2 + qt^5a^3bc^2d^2 + qt^5a^2bc^3d^2 + q^3t^6a^2bc + q^3t^6ab^2c + q^3t^6abc^2 + 2q^2t^5a^2b^2c^2 + q^3t^6a^2bd + q^3t^6a^2cd + 3q^3t^6abcd + q^3t^6ac^2d + q^3t^6bc^2d + \\
&3q^2t^5a^2bc^2d + qt^6a^2bc^2d + q^3t^6acd^2 + 2q^2t^5a^2c^2d^2 + qt^4a^2b^2c^2d^2 + q^3t^5abcd - qt^5a^2bc^2d + 2q^2t^5a^2bc + 2q^2t^5abc^2 + 2q^2t^5a^2cd - q^3t^4abcd + q^2t^5abcd - qt^4a^3bcd + 2q^2t^5ac^2d - qt^4abc^3d - 2q^2t^4a^2bc - \\
&2q^2t^4abc^2 - 2q^2t^4a^2cd + q^2t^4abcd - qt^3a^2b^2cd - 2q^2t^4ac^2d - qt^3ab^2c^2d - qt^3a^2bcd^2 - qt^3abc^2d^2 - qt^4a^2bc - q^2t^3ab^2c + qt^4abc^2 - q^2t^3a^2bd + qt^4a^2cd - 4q^2t^3abcd - qt^4abcd + qt^4ac^2d - q^2t^3bc^2d - t^3a^2bc^2d - \\
&q^2t^3acd^2 - q^2t^4a^2 - q^2t^4ab - q^2t^4ac - q^2t^4bc - 2qt^3a^2bc - q^2t^4c^2 - 2qt^3abc^2 - q^2t^4ad - q^2t^4cd - 2qt^3a^2cd + qt^3abcd - 2qt^3ac^2d - qt^4ac - 2qt^2abcd - qt^3a^2 - qt^3ac - qt^3c^2 - q^2t^3 + 2qt^2a^2 + 2qt^2ac + 2qt^2c^2 + \\
&qtab + qtbc + qtad + qtcd + qt^2 + tac - qt + q\big) z_3 + \tfrac{1}{(1-qt^2)(1-t^2abcd)(1-qt^3abcd)(1-q^2t^4abcd)} \big(q^4t^{11}a^3b^3c^3d + q^4t^{11}a^3b^2c^3d^2 + q^4t^{11}a^3bc^3d^3 + q^4t^{10}a^3b^3c^2d^2 + q^4t^{10}a^3b^2c^3d^2 + q^4t^{10}a^2b^3c^3d^2 + q^4t^{10}a^3b^2c^2d^3 + \\
&q^4t^{10}a^2b^2c^3d^3 + q^3t^{10}a^3b^3c^3d + q^3t^{10}a^3b^2c^3d^2 + q^3t^{10}a^3bc^3d^3 - q^4t^{10}a^3b^2c^2d - q^4t^{10}a^2b^3c^2d - q^4t^{10}a^2b^2c^3d - q^4t^{10}a^3b^2c^2d^2 - q^4t^{10}a^2b^3c^2d^2 - q^4t^{10}a^2b^2c^3d^2 - q^4t^{10}a^3bc^2d^3 - q^4t^{10}a^2bc^3d^3 + \\
&q^3t^{10}a^2b^2c^2d^3 + q^3t^{10}a^3b^3c^3d + q^3t^{10}a^3b^2c^3d^2 + q^3t^{10}a^3bc^3d^3 - q^4t^{10}a^3b^2c^2d - q^4t^{10}a^2b^3c^2d - q^4t^{10}a^2b^2c^3d - q^4t^{10}a^3b^2c^2d^2 - q^4t^{10}a^2b^3c^2d^2 - q^4t^{10}a^2b^2c^3d^2 - q^4t^{10}a^3bc^2d^3 - q^4t^{10}a^2bc^3d^3 + q^3t^{10}ab^2c^2d^2 + q^3t^{10}a^2b^3c^2d^2 + \\
&q^3t^9a^3bc^3d^3 - q^4t^9a^2b^2c^2d^2 - 2q^3t^9a^3b^2c^2d^2 - 2q^3t^9a^2b^2c^3d^2 - 2q^3t^9a^3bc^2d^3 - q^3t^7a^3b^3c^2d^2 - 2q^3t^9a^2bc^3d^2 - q^3t^7a^3b^2c^3d^2 - q^3t^7a^2b^3c^3d^2 + q^4t^9a^2b^2c^2d^2 - 2q^3t^8a^3b^3c^3d^2 + \\
&q^4t^9ab^2c^2d + q^3t^8a^3b^2c^2d - q^3t^8a^2b^3c^2d + q^3t^8a^3b^2c^3d^2 + q^3t^8a^2b^3c^3d^2 + q^3t^7a^3b^3c^3d^2 - q^4t^8a^2b^2c^2d^2 + q^3t^8a^2bc^3d^2 + q^3t^7a^2b^2c^3d^2 - q^4t^8a^2b^2c^2d - q^3t^8a^3b^2c^2d^2 - q^3t^7a^3b^3c^2d^2 - q^4t^8a^2b^2c^2d^2 - \\
&q^4t^8a^2bc^2d - q^4t^8ab^2c^2d - q^2t^8a^3b^2c^2d - q^2t^8a^2b^2c^3d - q^4t^8a^2bcd^2 + q^3t^7a^3b^2cd^2 - q^4t^8abc^2d^2 - q^2t^8a^3bc^2d^2 + q^3t^7ab^2c^3d^2 + q^3t^8a^3bcd + q^3t^8a^2b^2cd + 3q^3t^8a^2bc^2d + \\
&q^3t^8ab^2c^2d + q^2t^7a^3b^2c^2d + q^3t^8abc^3d + q^2t^7a^2b^2c^3d + q^3t^8a^2bcd^2 + q^3t^8abc^2d^2 + q^2t^7a^3bc^2d^2 + q^3t^6a^2b^2c^2d^2 + q^2t^7a^2bc^3d^2 - q^3t^7a^2b^2c^2 - q^4t^8abcd - q^3t^7a^2b^2cd - 5q^3t^7a^2bc^2d - q^3t^6ab^2c^2d + q^2t^5a^2b^3c^2d + \\
&q^2t^6a^3b^2c^2d + q^2t^6a^2b^2c^3d - q^3t^7a^2bcd^2 - q^3t^7a^2c^2d^2 - q^3t^7abc^2d^2 + q^2t^6a^3bc^2d^2 + q^4t^7abcd + q^2t^7a^3bcd - q^3t^6a^2b^2cd + 2q^3t^6a^2bc^2d + 3q^2t^7a^2bc^2d - q^3t^6ab^2c^2d + q^2t^5a^2b^3c^2d + \\
&q^2t^7abc^3d - q^3t^6a^2bcd^2 - q^3t^6abc^2d^2 + 2q^2t^5a^2b^2c^2d^2 + q^2t^5a^2bc^2d^3 - q^3t^7ab^2c - 2q^2t^6a^2b^2c^2 - 2q^3t^7abcd - q^2t^6a^3bcd + qt^5a^3b^2c^2d + 7q^2t^6a^2bc^2d + q^3t^5ab^2c^2d + qt^5a^3b^2c^2d - q^2t^6abc^3d + qt^5a^2b^3c^2d - \\
&q^3t^7acd^2 + q^3t^5a^2bcd^2 - 2q^2t^6a^2c^2d^2 + q^3t^5abc^2d^2 + qt^5a^3bc^2d^2 + qt^5a^2bc^3d^2 + q^3t^6a^2bc + q^3t^6ab^2c + q^3t^6abc^2 + 2q^2t^5a^2b^2c^2 + q^3t^6a^2bd + q^3t^6a^2cd + 4q^3t^6abcd - q^2t^5a^2b^2cd + q^3t^6ac^2d + q^3t^6bc^2d + \\
&4q^2t^5a^2bc^2d + qt^6a^2bc^2d - q^2t^5ab^2c^2d + q^3t^6acd^2 - q^2t^5a^2bcd^2 + 2q^2t^5a^2c^2d^2 - q^2t^5abc^2d^2 + q^3t^5abcd - q^2t^6abcd + q^2t^4a^2b^2cd - 3qt^5a^2bc^2d + q^2t^4ab^2c^2d + q^2t^4a^2bcd^2 + q^2t^4abc^2d^2 + 3q^2t^5a^2bc + \\
&q^2t^5ab^2c + 3q^2t^5abc^2 + qt^4a^2b^2c^2 + 3q^2t^5a^2cd - q^3t^4abcd + 3q^2t^5abcd - qt^4a^3bcd + 3q^2t^4a^2bc^2d + qt^4a^2c^2d - q^2t^4abc^2 - 2qt^4a^2bc^2 + q^2t^4a^2cd + 4qt^4abcd - qt^4ac^2d - q^2t^3bc^2d - t^3a^2bc^2d - \\
&q^2t^3acd^2 - q^2t^4a^2 - q^2t^4ab - q^2t^4ac - q^2t^4bc - 2qt^3a^2bc - q^2t^4c^2 - 2qt^3abc^2 - q^2t^4ad - q^2t^4cd - 2qt^3a^2cd + qt^3abcd - 2qt^3ac^2d - qt^4ac - 2qt^2abcd - qt^3a^2 - qt^3ac - qt^3c^2 - q^2t^3 + 2qt^2a^2 + 2qt^2ac + 2qt^2c^2 + \\
&qtab + qtbc + qtad + qtcd + tac - qt + q\big) z_2 - \tfrac{(1-t)qt^2}{(1-qt^2)(1-t^2abcd)(1-qt^3abcd)(1-q^2t^4abcd)} \big(q^2t^7a^3b^3c^3d + q^2t^7a^3b^2c^3d^2 + q^2t^7a^3bc^3d^3 + q^2t^6a^3b^2c^2d^2 + q^2t^6a^2b^2c^3d - q^2t^6a^2b^2c^2d - q^2t^6a^2b^3c^2d^2 - q^2t^6a^3b^2c^3d^2 - \\
&qt^5a^3b^2c^2d - qt^5a^2b^2c^3d - qt^5a^3bc^2d^2 - qt^5a^2bc^3d^2 + q^2t^4a^2bc^2d - q^2t^4a^2bc^3d^2 + qt^4abc^3d - qt^3a^2b^2c^2 - 2qt^3a^2bc^2d - t^2a^2bc^2d - qt^2a^2bc + qt^2abc^2 + \\
&qt^2a^2cd + qt^2ac^2d + ta^2bc + tabc^2 + ta^2cd + tac^2d - tac - a^2 - ac - c^2\big) z_1 + \tfrac{(1-t)t^2q^2(qt^2abc+qt^2acd-a-c)}{(1-qt^2)(1-q^2t^4abcd)} z_1^{-1} z_3 - \tfrac{t(1-t)q}{(1-qt^2)(1-t^2abcd)(1-q^2t^4abcd)} \big(q^2t^5a^2b^2c^2d + q^2t^5a^2bc^2d^2 - q^2t^5a^2bcd - q^2t^5abc^2d + q^2t^4abc + \\
&q^2t^4acd - q^2t^3abc - q^2t^3acd + qt^3abc + qt^3acd - qt^2abc - qt^2acd - qt^2a - qt^2c - tabc - tacd + qta + qtc + a + c\big) z_1^{-1} z_2 + \tfrac{tq(qt^2abc+qt^2acd-a-c)}{(1-q^2t^4abcd)} z_2^{-1} z_3 - \tfrac{t}{(1-qt^2)(1-t^2abcd)(1-qt^3abcd)(1-q^2t^4abcd)} \big(q^4t^9a^3b^3c^3d^2 + \\
&q^4t^9a^3b^2c^3d^3 + q^4t^9a^3b^3c^2d + q^4t^9a^3b^2c^3d + q^4t^9a^2b^3c^3d + q^4t^9a^3bc^3d^2 + q^4t^9a^2b^3c^2d^2 + q^4t^9a^2b^2c^3d^2 + q^4t^9a^2bc^3d^3 + q^3t^9a^3b^3c^3 + q^3t^9a^3b^2c^3d + 2q^4t^8a^3b^2c^2d^2 + q^3t^9a^3bc^3d^2 + 2q^4t^8a^2b^2c^3d^2 - q^3t^7a^3b^3c^3d^2 + \\
&q^3t^9a^3c^3d^3 - q^3t^7a^3b^2c^3d^3 + q^3t^8a^3b^2c^3d - q^4t^7a^2b^2c^2d - q^4t^8a^3bc^2d - q^4t^8a^2b^2c^2d - q^4t^8a^2bc^3d^2 - q^4t^8ab^2c^3d^2 - q^4t^8a^2b^3c^2d - q^4t^8a^2b^2c^3d - q^3t^8a^3bcd^2 - q^3t^7a^3b^2c^2d^2 - \\
&q^4t^8abc^3d^2 + q^3t^7a^2b^2c^3d^2 - q^3t^8a^3b^2c^2 - 3q^4t^7a^2b^2c^2d - q^3t^6a^3b^3c^2d - q^3t^8a^2bc^3d - q^3t^6a^3b^2c^3d - q^3t^6a^2b^3c^3d - q^3t^8a^3c^2d^2 - 3q^4t^7a^2bc^2d^2 - 2q^3t^6a^3bc^2d^2 - \\
&q^3t^6a^2b^3c^2d^2 - 2q^3t^6a^2b^2c^3d^2 - q^3t^6a^3bc^2d^3 - q^3t^6a^2b^2c^2d^3 - q^3t^7a^2b^3c^2 - q^3t^7a^2b^2c^3 - q^2t^6a^3b^3c^3 - q^3t^7a^3b^2cd - 2q^3t^7a^3bc^2d + q^4t^6a^2b^2c^2d - 2q^3t^7a^2b^2c^2d - 2q^3t^7a^2bc^3d + \\
&q^2t^6a^3b^2c^3d - q^3t^7a^3bcd^2 - q^3t^7a^2c^2d^2 + q^4t^6a^2bc^2d^2 - 2q^3t^7a^2bc^2d^2 - q^3t^7a^2c^3d^2 - q^3t^7abc^3d^2 - q^2t^6a^3b^2c^2d^2 + q^2t^6a^2bc^3d^2 - q^2t^7a^3b^2c^2d - q^2t^7a^2b^3c^2d + q^4t^7abcd + q^3t^7a^3bcd + \\
&5q^3t^6a^2b^2c^2d - q^2t^7a^2bc^3d - q^2t^5a^3b^2c^3d - q^2t^7a^3c^2d^2 - 5q^3t^6a^2bc^2d^2 - q^2t^7a^2c^3d^2 - q^2t^5a^3bc^3d^2 + q^3t^7a^2bc^2 + q^3t^5a^3b^2cd + q^3t^7a^2c^2d + q^3t^5a^2b^3cd - q^2t^6a^3bc^2d + 5q^3t^5a^2b^2c^2d + q^3t^5a^2bc^3d - \\
&q^2t^6a^2bc^3d + q^3t^5ab^2c^3d + q^3t^5a^3bcd^2 + 5q^3t^5a^2bc^2d^2 + q^3t^5abc^3d^2 + q^3t^6a^3bc + q^3t^6a^2b^2c + q^3t^6a^2bc^2 + q^3t^6ab^2c^2 + q^2t^5a^3b^2c^2 + q^3t^6abc^3 + q^2t^5a^2b^2c^3 + q^3t^6a^3bd + q^3t^6a^3cd + 4q^3t^6a^2bcd + q^3t^6a^2c^2d + \\
&4q^3t^6abc^2d + q^2t^5a^3bc^2d - 2q^2t^5a^2b^2c^2d + q^3t^6ac^3d + q^3t^6bc^3d + q^2t^5a^2bc^3d + q^3t^6ac^2d^2 + q^2t^5a^3c^2d^2 + q^2t^6a^3bc + 2q^2t^6a^2b^2c^2 + q^2t^4a^3b^2c^2 + q^2t^4a^2b^3c^2 + \\
&q^2t^6abc^3 + q^2t^4a^2b^2c^3 + q^2t^6a^3cd + q^3t^5a^2bcd + q^2t^4a^3b^2cd + 2q^2t^6a^2c^2d + q^3t^5abc^2d + 2q^2t^4a^3bc^2d + 6q^2t^4a^2b^2c^2d + q^2t^6ac^3d + 2q^2t^4a^2bc^3d + q^2t^4ab^2c^3d + q^2t^4a^3bcd^2 + q^2t^4a^3c^2d^2 + 6q^2t^4a^2bc^2d^2 + \\
&q^2t^4a^2c^3d^2 + q^2t^4abc^3d^2 + q^2t^4a^2c^2d^3 + q^2t^5a^3bc + q^2t^5a^2b^2c + q^2t^5a^2bc^2 + q^2t^5ab^2c^2 + qt^4a^3b^2c^2 + q^2t^5abc^3 + qt^4a^2b^2c^3 + q^2t^5a^3cd + q^2t^4a^2bcd + 2q^2t^5a^2bcd + q^2t^5a^2c^2d - q^3t^4abc^2d + 2q^2t^5abc^2d + \\
&qt^4a^3bc^2d + q^2t^5ac^3d + qt^4a^2bc^3d + q^2t^5a^2cd^2 + q^2t^5ac^2d^2 + qt^4a^3c^2d^2 + qt^4a^2c^3d^2 - 2q^3t^5abc - q^2t^4a^2bc^2 + qt^5a^2bc^2 - 2q^3t^5acd + q^2t^4a^2bcd - q^2t^4a^2c^2d + qt^5a^2c^2d + q^2t^4abc^2d + qt^3a^3bc^2d + \\
&qt^3a^2b^2c^2d + qt^3a^2bc^3d + qt^3a^2bc^2d^2 - q^2t^5a^2c + 3q^2t^4abc - q^2t^5a^3bc - q^2t^5a^2b^2c - q^2t^5ac^2 - q^2t^3a^2bc^2 - q^2t^3ab^2c^2 - q^2t^3abc^3 - q^2t^3a^3bd + 3q^2t^4acd - q^2t^3a^3cd - 5q^2t^3a^2bcd - q^2t^3a^2c^2d - 5q^2t^3abc^2d - \\
&q^2t^3ac^3d - q^2t^3bc^3d - q^2t^3a^2cd^2 - q^2t^3acd^2 - q^2t^4a^3 - q^2t^4a^2b - q^2t^4a^2c - q^3t^3abc - 3q^2t^4abc - qt^3a^3bc - q^2t^4ac^2 - q^2t^4bc^2 - 2qt^3a^2bc^2 - q^2t^4c^3 - qt^3abc^3 - q^2t^4a^2d - q^3t^3acd - 3q^2t^4acd - qt^3a^3cd - \\
&q^2t^4c^2d - 2qt^3a^2c^2d - qt^3ac^3d - qt^4a^2c + 4q^2t^3abc - qt^2a^3bc - qt^2a^2b^2c - qt^4ac^2 - qt^2a^2bc^2 - qt^2ab^2c^2 - qt^2abc^3 + 4q^2t^3acd - qt^2a^3cd - 3qt^2a^2bcd - qt^2a^2c^2d - 3qt^2abc^2d - qt^2ac^3d - qt^2a^2cd^2 - \\
&qt^2ac^2d^2 - 2q^2t^2abc - qt^3abc - t^2a^2bc^2 - 2q^2t^2acd - qt^3acd - t^2a^2c^2d + q^2t^3a + q^2t^3c + qt^2a^2c + 2qt^2abc + qt^2ac^2 + 2qt^2acd - 3q^2t^2a + qta^3 + qta^2b - 3q^2t^2c + qta^2c + qtac^2 + qtbc^2 + qtc^3 + qta^2d + qtc^2d + \\
&q^2ta + qt^2a + q^2tc + qt^2c + ta^2c + tac^2 - 2qta - 2qtc + 2qa + 2qc\big) - \tfrac{t(1-t)q(qt^3a^2bcd+qt^3abc^2d-qt^2abc-qt^2acd-tabc-tacd+a+c)}{(1-t^2abcd)(1-q^2t^4abcd)} z_2 z_3^{-1} - \tfrac{(1-t)t^2q^2}{(1-qt^2)(1-t^2abcd)(1-qt^3abcd)(1-q^2t^4abcd)} \big(q^2t^7a^3b^3c^3d + q^2t^7a^3b^2c^3d^2 + \\
&q^2t^7a^3bc^3d^3 + q^2t^6a^3b^2c^3d^2 - q^2t^6a^3b^2c^2d - q^2t^6a^2b^2c^3d - q^2t^6a^3bc^2d^2 - q^2t^6a^2b^2c^2d^2 - q^2t^6a^2bc^3d^2 - qt^5a^3bc^2d^2 - qt^5a^2bc^3d^2 + q^2t^5a^2bc^2d - q^2t^4a^2bc^2d + qt^4a^3bcd + 2qt^4a^2bc^2d + qt^4abc^3d - \\
&qt^3a^2b^2c^2 - 2qt^3a^2bc^2d - qt^3a^2c^2d^2 + t^3a^2bc^2d - t^2a^2bc^2d + qt^2a^2bc + qt^2abc^2 + qt^2a^2cd + qt^2ac^2d + ta^2bc + tabc^2 + ta^2cd + tac^2d - a^2 - ac - c^2\big) z_3^{-1}
\end{aligned}$$

The affine Weyl group of type $\widetilde{C}_n$

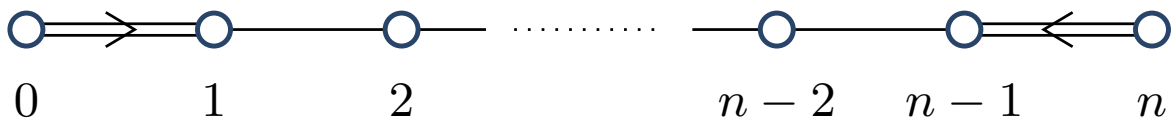
The affine Weyl group W of type $\widetilde{C}_n$ is generated by $s_0, s_1, \ldots, s_n$ with relations

$$\begin{aligned} s_i s_j &= s_j s_i, && |i-j| \geq 2, \\ s_i s_{i+1} s_i &= s_{i+1} s_i s_{i+1}, && 1 \leq i < n-1, \\ s_0 s_1 s_0 s_1 &= s_1 s_0 s_1 s_0, \\ s_{n-1} s_n s_{n-1} s_n &= s_n s_{n-1} s_n s_{n-1}, \\ s_i^2 &= 1, && 0 \leq i \leq n. \end{aligned}$$



The finite Weyl group $W_0 = \langle s_1, \ldots, s_n \rangle \simeq S_n \ltimes (\mathbb{Z}/2\mathbb{Z})^n$ is the group of signed permutations.

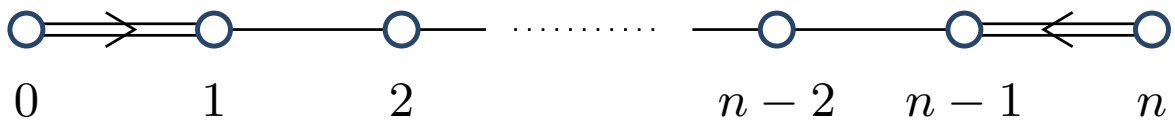
The affine Weyl group of type $\widetilde{C}_n$



W acts on Laurent polynomials in $z_1, \ldots, z_n$ by

$$
\begin{aligned}
s_i(z_1, \ldots, z_i, z_{i+1}, \ldots, z_n) &= (z_1, \ldots, z_{i+1}, z_i, \ldots, z_n), \qquad 1 \le i < n, \\
s_n(z_1, \ldots, z_{n-1}, z_n) &= (z_1, \ldots, z_{n-1}, z_n^{-1}), \\
s_0(z_1, z_2, \ldots, z_n) &= (q z_1^{-1}, z_2, \ldots, z_n).
\end{aligned}
$$

The affine Weyl group of type $\widetilde{C}_n$



W acts on Laurent polynomials in $z_1, \ldots, z_n$ by

$$\begin{aligned} s_i(z_1, \ldots, z_i, z_{i+1}, \ldots, z_n) &= (z_1, \ldots, z_{i+1}, z_i, \ldots, z_n), \qquad 1 \le i < n, \\ s_n(z_1, \ldots, z_{n-1}, z_n) &= (z_1, \ldots, z_{n-1}, z_n^{-1}), \\ s_0(z_1, z_2, \ldots, z_n) &= (qz_1^{-1}, z_2, \ldots, z_n). \end{aligned}$$

A Laurent polynomial is symmetric if it is invariant under the action of W_0.

Example: The symmetric Koornwinder polynomial $K_{(2,0)}$

$$\begin{aligned} K_{(2,0)} = z_1^2 + z_1^{-2} + z_2^2 + z_2^{-2} + \frac{(1+q)(1-t)}{1-qt}(z_1 + z_1^{-1})(z_2 + z_2^{-1}) \\ - \frac{(1+q)(1-qt^2)}{(1-qt)(1-q^2t^2abcd)}\big(a+b+c+d-qt(abc+abd+acd+bcd)\big) \\ \times (z_1 + z_1^{-1} + z_2 + z_2^{-1}) + \text{constant}. \end{aligned}$$

The affine Hecke algebra $\mathcal{H}$ of type $\widetilde{C}_n$

The affine Hecke algebra $\mathcal{H}$ is generated by $T_0, T_1, \ldots, T_n$, subject to the braid relations

$$\begin{aligned} T_i T_j &= T_j T_i, && |i-j| \geq 2, \\ T_i T_{i+1} T_i &= T_{i+1} T_i T_{i+1}, && 1 \leq i < n-1, \\ T_0 T_1 T_0 T_1 &= T_1 T_0 T_1 T_0, \\ T_{n-1} T_n T_{n-1} T_n &= T_n T_{n-1} T_n T_{n-1}, \end{aligned}$$

and the quadratic relations

$$(T_i - t_i)(T_i + 1) = 0, \qquad 0 \leq i \leq n,$$

where $t_0 = -ac/q$, $t_i = t$ for $1 \leq i < n$, and $t_n = -bd$.



When $t_i = 1$ for all $0 \leq i \leq n$, we get back the affine Weyl group W, where T_i corresponds to s_i.

Polynomial representation of $\mathcal{H}$

The affine Hecke algebra $\mathcal{H}$ acts on Laurent polynomials in $z_1, \ldots, z_n$ by the Demazure–Lusztig operators

$$T_i = t + \frac{tz_i - z_{i+1}}{z_i - z_{i+1}}\,(s_i - 1), \qquad 1 \le i < n,$$

$$T_n = -bd + \frac{(1 - bz_n)(1 - dz_n)}{1 - z_n^2}\,(s_n - 1),$$

$$T_0 = -\frac{ac}{q} + \frac{(1 - a/z_1)(1 - c/z_1)}{1 - q/z_1^2}\,(s_0 - 1).$$

Theorem (Sahi '99)

This representation is faithful.

Koornwinder ASEP polynomials

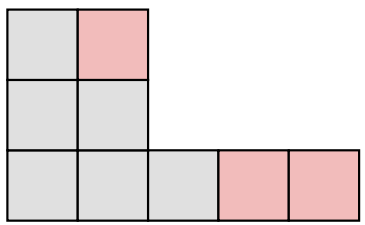
Fix a partition λ. The Koornwinder ASEP polynomials $\{F_\eta\}_{\eta \in W_0 \cdot \lambda}$ are the unique family of Laurent polynomials, normalized by $[z^\eta]F_\eta = 1$, satisfying the qKZ relations (Kasatani '11)

$$
\begin{aligned}
T_0 F_{\eta_1,\eta_2,\ldots,\eta_n} &= q^{\eta_1} F_{-\eta_1,\eta_2,\ldots,\eta_n}, && \eta_1 < 0, \\
T_0 F_{\eta_1,\eta_2,\ldots,\eta_n} &= t_0\, F_{\eta_1,\eta_2,\ldots,\eta_n}, && \eta_1 = 0, \\
T_i F_{\ldots,\eta_i,\eta_{i+1},\ldots} &= t\, F_{\ldots,\eta_i,\eta_{i+1},\ldots}, && \eta_i = \eta_{i+1},\ 1 \le i < n, \\
T_i F_{\ldots,\eta_i,\eta_{i+1},\ldots} &= F_{\ldots,\eta_{i+1},\eta_i,\ldots}, && \eta_i > \eta_{i+1},\ 1 \le i < n, \\
T_n F_{\ldots,\eta_n} &= t_n\, F_{\ldots,\eta_n}, && \eta_n = 0, \\
T_n F_{\ldots,\eta_n} &= F_{\ldots,-\eta_n}, && \eta_n > 0.
\end{aligned}
$$

Recall that $t_0 = -ac/q$ and $t_n = -bd$.

Branching rules for Schur polynomials

$$s_\lambda(z_1, \ldots, z_n) = \sum_{\substack{\mu \subseteq \lambda \\ \lambda/\mu \text{ horizontal strip}}} z_n^{|\lambda|-|\mu|}\, s_\mu(z_1, \ldots, z_{n-1}).$$



$\lambda = (5, 2, 2)$, $\mu = (3, 2, 1)$

λ/μ is a horizontal strip: at most one box in each column.

Branching rules for Macdonald polynomials

$$P_\lambda(z_1,\dots,z_n;q,t) = \sum_{\substack{\mu\subseteq\lambda \\ \lambda/\mu \text{ horizontal strip}}} \psi_{\lambda/\mu}(q,t)\, z_n^{|\lambda|-|\mu|} P_\mu(z_1,\dots,z_{n-1};q,t),$$

with branching coefficients (Macdonald '95)

$$\psi_{\lambda/\mu}(q,t) = \prod_{s\in R_{\lambda/\mu}\setminus C_{\lambda/\mu}} \frac{b_\mu(s)}{b_\lambda(s)}, \qquad b_\lambda(s) = \frac{1-q^{a_\lambda(s)}t^{l_\lambda(s)+1}}{1-q^{a_\lambda(s)+1}t^{l_\lambda(s)}},$$

where $R_{\lambda/\mu}$ (resp. $C_{\lambda/\mu}$) is the union of rows (resp. columns) meeting λ/μ.

Branching rules for Macdonald polynomials

$$P_\lambda(z_1,\dots,z_n;q,t) = \sum_{\substack{\mu\subseteq\lambda \\ \lambda/\mu \text{ horizontal strip}}} \psi_{\lambda/\mu}(q,t)\, z_n^{|\lambda|-|\mu|} P_\mu(z_1,\dots,z_{n-1};q,t),$$

with branching coefficients (Macdonald '95)

$$\psi_{\lambda/\mu}(q,t) = \prod_{s\in R_{\lambda/\mu}\setminus C_{\lambda/\mu}} \frac{b_\mu(s)}{b_\lambda(s)}, \qquad b_\lambda(s) = \frac{1-q^{a_\lambda(s)}t^{l_\lambda(s)+1}}{1-q^{a_\lambda(s)+1}t^{l_\lambda(s)}},$$

where $R_{\lambda/\mu}$ (resp. $C_{\lambda/\mu}$) is the union of rows (resp. columns) meeting λ/μ.

Branching rules also exist for

- interpolation Macdonald polynomials (Okounkov '98),
- symmetric Koornwinder polynomials (van Diejen–Emsiz '15).

Branching rules for Macdonald polynomials

$$P_\lambda(z_1,\dots,z_n;q,t) = \sum_{\substack{\mu\subseteq\lambda \\ \lambda/\mu \text{ horizontal strip}}} \psi_{\lambda/\mu}(q,t)\, z_n^{|\lambda|-|\mu|} P_\mu(z_1,\dots,z_{n-1};q,t),$$

with branching coefficients (Macdonald '95)

$$\psi_{\lambda/\mu}(q,t) = \prod_{s\in R_{\lambda/\mu}\setminus C_{\lambda/\mu}} \frac{b_\mu(s)}{b_\lambda(s)}, \qquad b_\lambda(s) = \frac{1-q^{a_\lambda(s)}t^{l_\lambda(s)+1}}{1-q^{a_\lambda(s)+1}t^{l_\lambda(s)}},$$

where $R_{\lambda/\mu}$ (resp. $C_{\lambda/\mu}$) is the union of rows (resp. columns) meeting λ/μ.

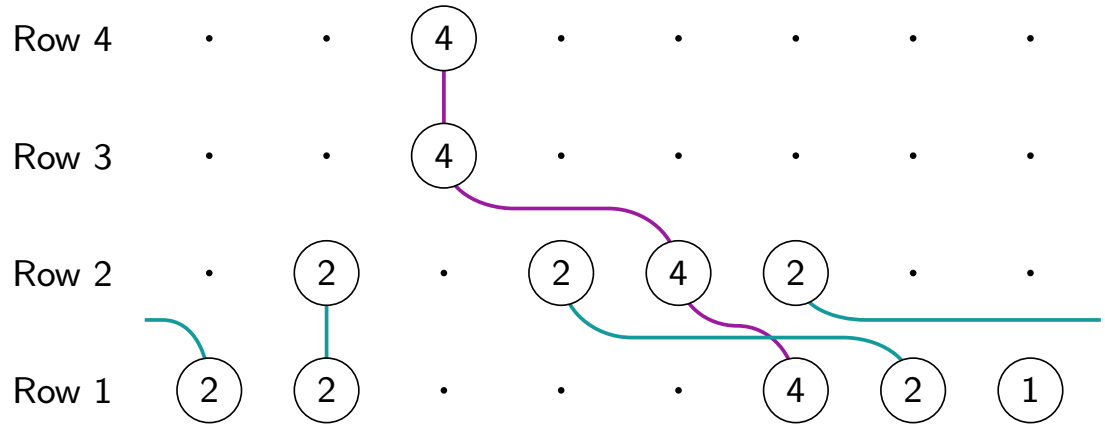
Branching rules also exist for

- interpolation Macdonald polynomials (Okounkov '98),
- symmetric Koornwinder polynomials (van Diejen–Emsiz '15).

These branching rules have two limitations:

- they do not have a direct analogue for nonsymmetric polynomials;
- except for the Schur case, the branching coefficients are complicated rational expressions in q and t, and the resulting formulas do not make integrality and positivity manifest.

Type A: multiline queues



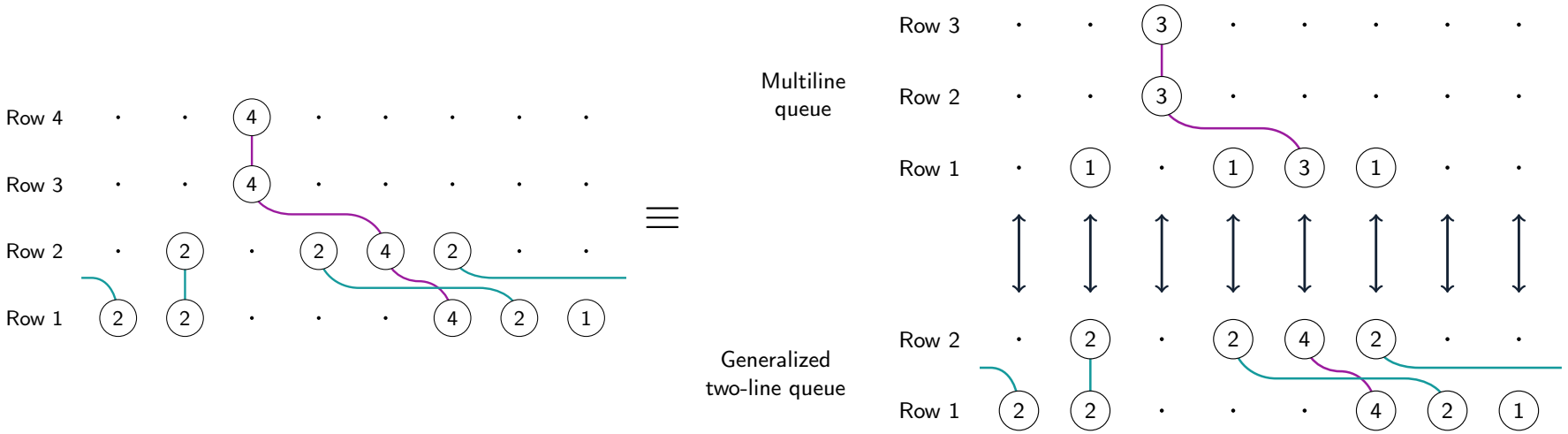
A multiline queue (MLQ) with 4 rows, of type $(2, 2, 0, 0, 0, 4, 2, 1)$.

- The type of an MLQ is given by the labels of its bottom row (0 for an empty position).
- Ball weight: each ball in column i contributes z_i; here $z_1 z_2^2 z_3^2 z_4 z_5 z_6^2 z_7 z_8$.
- Pairing weight: a rational function in q, t, multiplicative over the layers: here $\mathrm{wt}_{\mathrm{pairing}} = \mathrm{wt}_1\, \mathrm{wt}_2\, \mathrm{wt}_3$.

Theorem (Corteel–Mandelshtam–Williams '22)

For $\kappa \in \mathbb{Z}^n_{\geq 0}$, F^A_κ is the generating function of MLQs of type κ.

Type A two-row recurrence

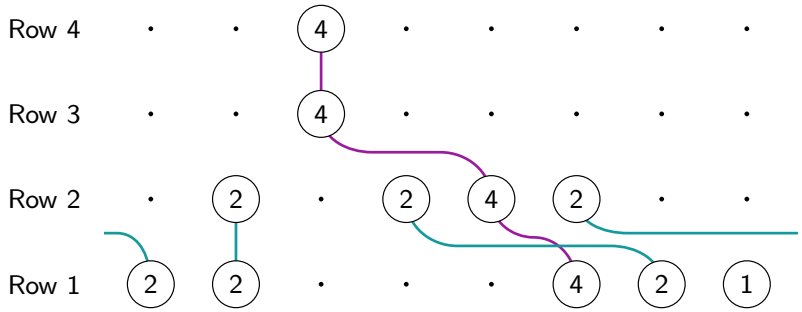


Since the pairing weight is multiplicative over layers, removing the bottom layer gives

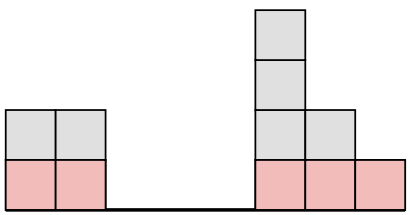
$$F_\kappa^A(z_1,\dots,z_n;q,t) = \prod_{i:\,\kappa_i>0} z_i \sum_{\mu\in S_n\cdot\kappa^-} c_\kappa^\mu(q,t)\, F_\mu^A(z_1,\dots,z_n;q,t),$$

where $\kappa_i^- = \max\{\kappa_i - 1, 0\}$, and $c_\kappa^\mu(q,t)$ is the generating function of generalized two-line queues.

Type A two-row recurrence

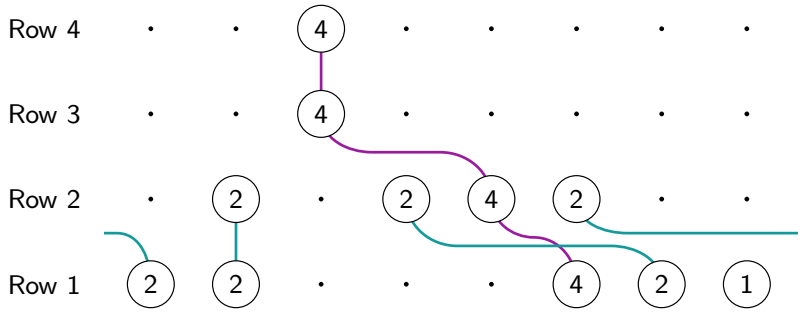


$$F_\kappa^A(z_1,\ldots,z_n;q,t) = \prod_{i:\,\kappa_i>0} z_i \sum_{\mu\in S_n\cdot\kappa^-} c_\kappa^\mu(q,t)\, F_\mu^A(z_1,\ldots,z_n;q,t)$$

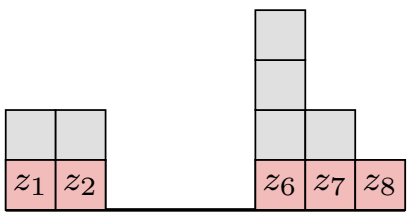


$\kappa = (2,2,0,0,0,4,2,1)$

Type A two-row recurrence

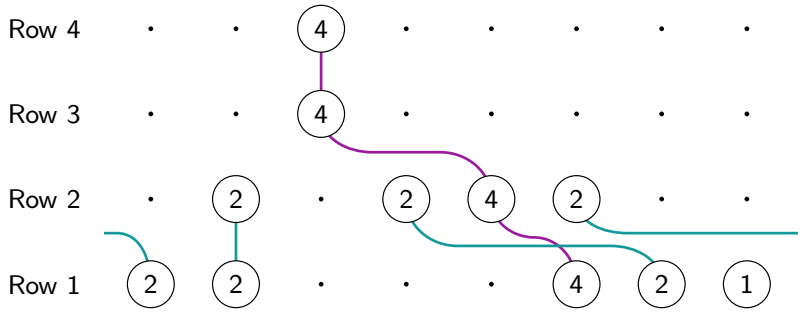


$$F_\kappa^A(z_1,\ldots,z_n;q,t) = \prod_{i:\,\kappa_i>0} z_i \sum_{\mu\in S_n\cdot\kappa^-} c_\kappa^\mu(q,t)\, F_\mu^A(z_1,\ldots,z_n;q,t)$$

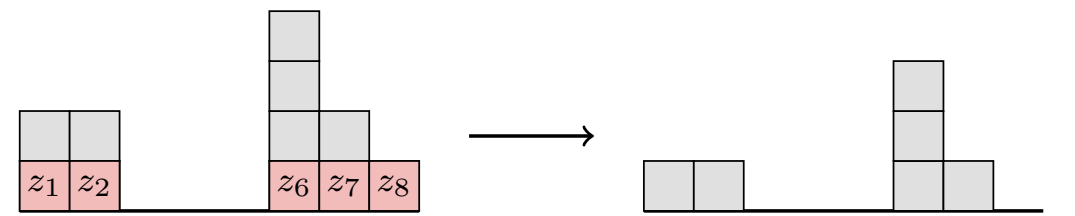


$\kappa = (2,2,0,0,0,4,2,1)$

Type A two-row recurrence

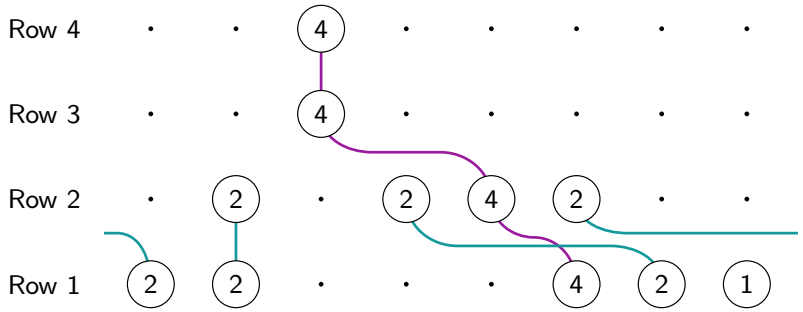


$$F_\kappa^A(z_1,\ldots,z_n;q,t) = \prod_{i:\,\kappa_i>0} z_i \sum_{\mu\in S_n\cdot\kappa^-} c_\kappa^\mu(q,t)\, F_\mu^A(z_1,\ldots,z_n;q,t)$$

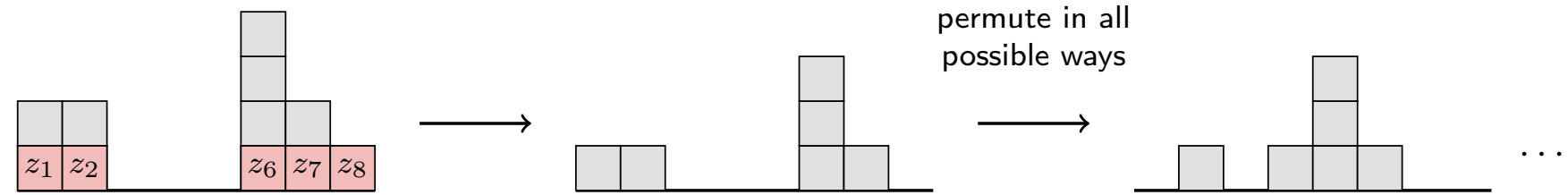


$\kappa = (2,2,0,0,0,4,2,1)$ $\qquad$ $\kappa^- = (1,1,0,0,0,3,1,0)$

Type A two-row recurrence



$$F_\kappa^A(z_1,\ldots,z_n;q,t) = \prod_{i:\,\kappa_i>0} z_i \sum_{\mu\in S_n\cdot\kappa^-} c_\kappa^\mu(q,t)\, F_\mu^A(z_1,\ldots,z_n;q,t)$$



Two-row recurrence for Koornwinder polynomials

Theorem A (two-row recurrence)

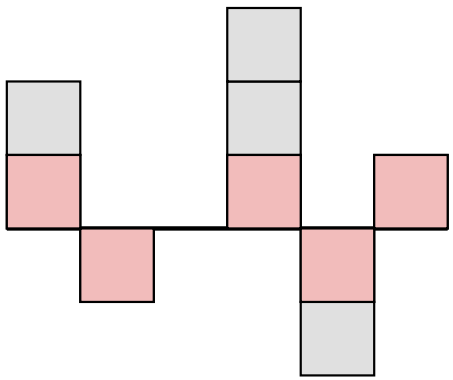
For $\kappa \in \mathbb{Z}^n$, *set* $(\kappa^-)_i = \kappa_i - \mathrm{Sign}(\kappa_i)$. *Then*

$$F_\kappa(z_1, \ldots, z_n; q, t, a, b, c, d) = \sum_{(\nu,\mu)} a_\kappa^\mu(\nu)\, z^\nu F_\mu(z_1, \ldots, z_n; q, t, a, b, c, d),$$

where $\mu \in W_0 \cdot \kappa^-$ *and* $\nu \in \mathbb{Z}^n$ *with* $\nu_i \in \{0, \mathrm{Sign}(\kappa_i)\}$.

Two-row recurrence for Koornwinder polynomials

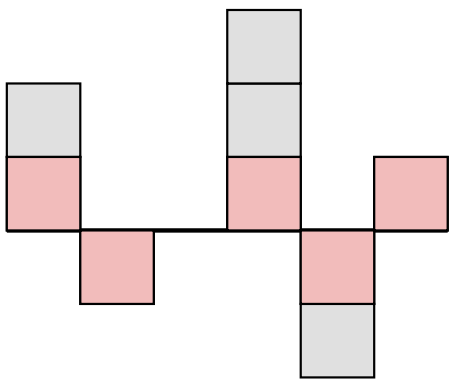
$$F_\kappa(z_1,\dots,z_n;q,t,a,b,c,d) = \sum_{(\nu,\mu)} a_\kappa^\mu(\nu)\, z^\nu F_\mu(z_1,\dots,z_n;q,t,a,b,c,d)$$



$\kappa = (2,-1,0,3,-2,1)$

Two-row recurrence for Koornwinder polynomials

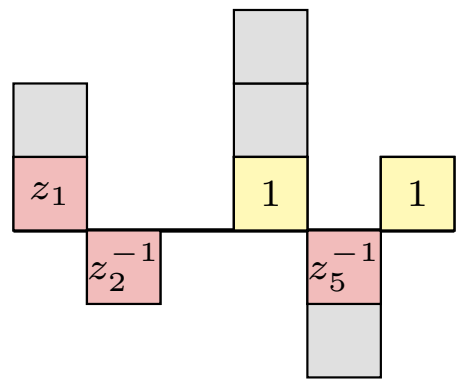
$$F_\kappa(z_1,\dots,z_n;q,t,a,b,c,d) = \sum_{(\nu,\mu)} a_\kappa^\mu(\nu)\, z^\nu F_\mu(z_1,\dots,z_n;q,t,a,b,c,d)$$



$\kappa = (2,-1,0,3,-2,1)$

for each basement cell, choose $z_i^{\mathrm{Sign}(\kappa_i)}$ or 1 according to ν

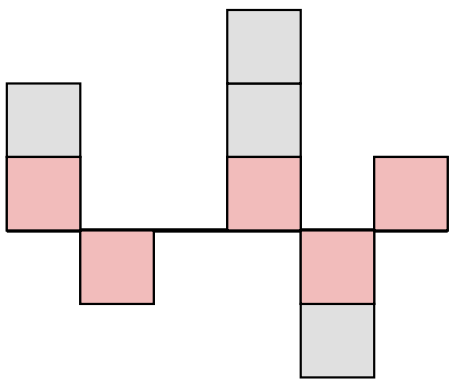
$\nu = (1,-1,0,0,-1,0)$



$z^\nu = z_1 z_2^{-1} z_5^{-1}$

Two-row recurrence for Koornwinder polynomials

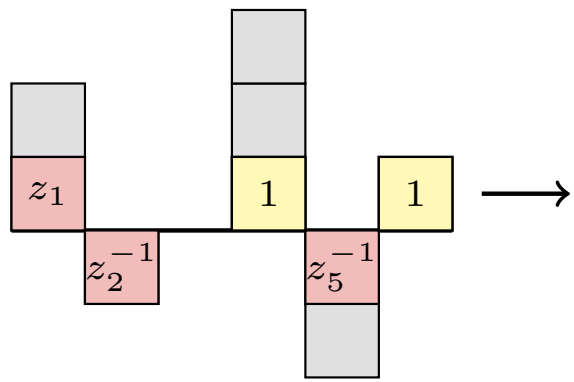
$$F_\kappa(z_1,\dots,z_n;q,t,a,b,c,d) = \sum_{(\nu,\mu)} a_\kappa^\mu(\nu)\, z^\nu F_\mu(z_1,\dots,z_n;q,t,a,b,c,d)$$



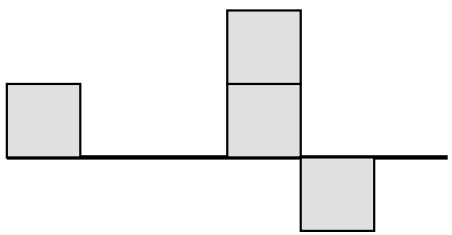
$\kappa = (2,-1,0,3,-2,1)$

for each basement cell, choose $z_i^{\mathrm{Sign}(\kappa_i)}$ or 1 according to ν

$\nu = (1,-1,0,0,-1,0)$



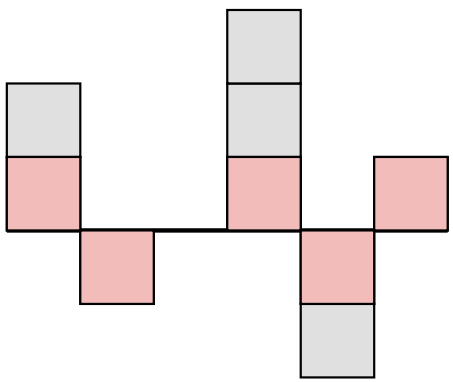
$z^\nu = z_1 z_2^{-1} z_5^{-1}$



$\kappa^- = (1,0,0,2,-1,0)$

Two-row recurrence for Koornwinder polynomials

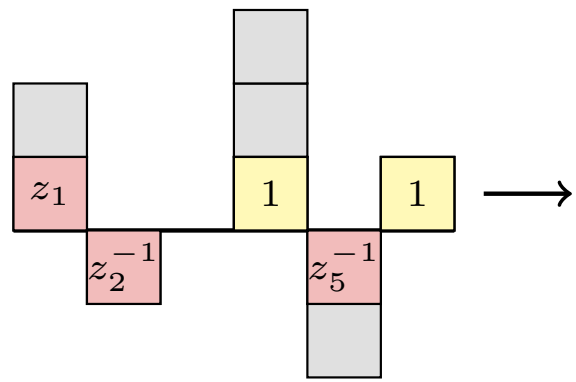
$$F_\kappa(z_1,\dots,z_n;q,t,a,b,c,d) = \sum_{(\nu,\mu)} a_\kappa^\mu(\nu)\, z^\nu F_\mu(z_1,\dots,z_n;q,t,a,b,c,d)$$



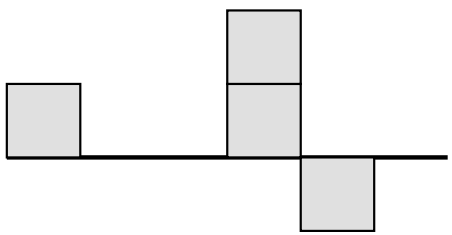
$\kappa = (2,-1,0,3,-2,1)$

for each basement cell, choose $z_i^{\mathrm{Sign}(\kappa_i)}$ or 1 according to ν

$\nu = (1,-1,0,0,-1,0)$

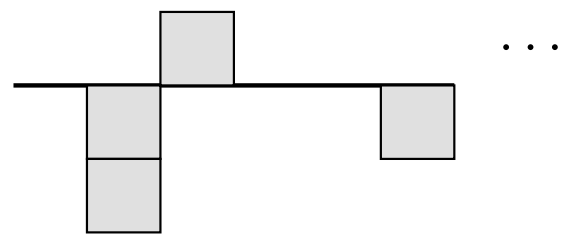


$z^\nu = z_1 z_2^{-1} z_5^{-1}$



$\kappa^- = (1,0,0,2,-1,0)$

permute and flip signs in all possible ways



$\dots$

$\mu = (0,-2,1,0,0,-1)$

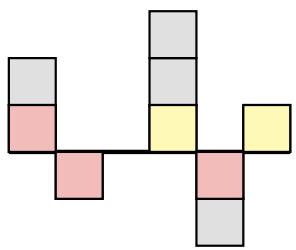
Two-line configurations

For $\kappa \in \mathbb{Z}^n$, $\nu \in \mathbb{Z}^n$ with $\nu_i \in \{0, \mathrm{Sign}(\kappa_i)\}$, and $\mu \in W_0 \cdot \kappa^-$, we write

$$a_\kappa^\mu(\nu) = \left[\begin{array}{c} \sigma \\ \tau \end{array} \right].$$

τ is obtained from κ by putting a bar on κ_i when $\nu_i = 0$.

Example: $\kappa = (2, -1, 0, 3, -2, 1)$, $\nu = (1, -1, 0, 0, -1, 0)$.



$$\tau = (2, -1, 0, \overline{3}, -2, \overline{1})$$

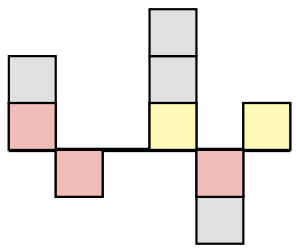
Two-line configurations

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$$a_\kappa^\mu(\nu) = \begin{bmatrix} \sigma \\ \tau \end{bmatrix}.$$

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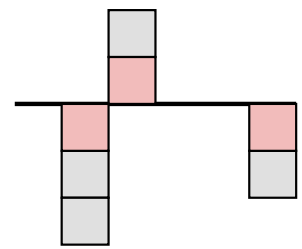
Example: $\kappa = (2, -1, 0, 3, -2, 1)$, $\nu = (1, -1, 0, 0, -1, 0)$.



$\tau = (2, -1, 0, \overline{3}, -2, \overline{1})$

$\sigma = \mu \oplus \mathrm{Sign}(\mu)$ is obtained from μ by adding a basement cell to each nonzero column.

Example: $\mu = (0, -2, 1, 0, 0, -1)$.



$\sigma = (0, -3, 2, 0, 0, -2)$

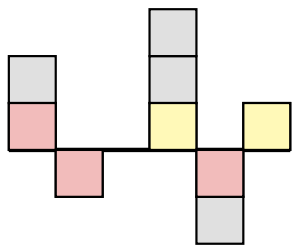
Two-line configurations

For $\kappa \in \mathbb{Z}^n$, $\nu \in \mathbb{Z}^n$ with $\nu_i \in \{0, \mathrm{Sign}(\kappa_i)\}$, and $\mu \in W_0 \cdot \kappa^-$, we write

$$a_\kappa^\mu(\nu) = \begin{bmatrix} \sigma \\ \tau \end{bmatrix}.$$

τ is obtained from κ by putting a bar on κ_i when $\nu_i = 0$.

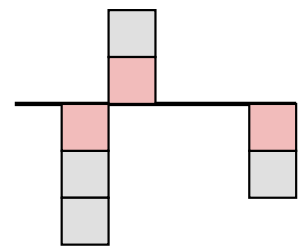
Example: $\kappa = (2, -1, 0, 3, -2, 1)$, $\nu = (1, -1, 0, 0, -1, 0)$.



$$\tau = (2, -1, 0, \overline{3}, -2, \overline{1})$$

$\sigma = \mu \oplus \mathrm{Sign}(\mu)$ is obtained from μ by adding a basement cell to each nonzero column.

Example: $\mu = (0, -2, 1, 0, 0, -1)$.



$$\sigma = (0, -3, 2, 0, 0, -2)$$

$$a_\kappa^\mu(\nu) = \begin{bmatrix} 0 & -3 & 2 & 0 & 0 & -2 \\ 2 & -1 & 0 & \overline{3} & -2 & \overline{1} \end{bmatrix}$$

Two-line configurations

Fix a partition λ. The recurrence coefficients are encoded by the diagrams

$$\begin{bmatrix} \sigma \\ \tau \end{bmatrix},$$

where

- τ is a signed permutation of λ, potentially with some bars,
- σ is a signed permutation of λ with its parts equal to 1 replaced by 0.

Example: for $\lambda = (3, 2, 2, 1, 1, 0)$,

$$\begin{bmatrix} 0 & -3 & 2 & 0 & 0 & -2 \\ 2 & -1 & 0 & \overline{3} & -2 & \overline{1} \end{bmatrix}.$$

Local rules

Theorem B

The two-line configurations are characterized by a family of local rules.

Only the columns involved are represented; the other columns are the same on both sides. Here $m > \ell$ and $k > 0$.

In the bulk (two consecutive columns):

$$\begin{bmatrix} m & \ell \\ -k & 0 \end{bmatrix} = t \begin{bmatrix} \ell & m \\ 0 & -k \end{bmatrix} + (1-t) \begin{bmatrix} m & \ell \\ 0 & -k \end{bmatrix}.$$

On the boundary (first column):

$$\begin{bmatrix} 0 & \cdots \\ \overline{k} & \cdots \end{bmatrix} = -q^{k-1}ac \begin{bmatrix} 0 & \cdots \\ \overline{-k} & \cdots \end{bmatrix} - q^{k-1}(a+c) \begin{bmatrix} 0 & \cdots \\ -k & \cdots \end{bmatrix}.$$

In total, there are around 90 local rules (43 up to symmetry).

A cyclic algorithm

Each local rule transports an active column to the next position, plus smaller terms, for some order on two-line configurations.

$$\begin{bmatrix} m & \ell \\ -k & 0 \end{bmatrix} = t \begin{bmatrix} \ell & m \\ 0 & -k \end{bmatrix} + \underbrace{(1-t)\begin{bmatrix} m & \ell \\ 0 & -k \end{bmatrix}}_{\text{smaller terms}},$$

$$\begin{bmatrix} 0 & \cdots \\ \overline{k} & \cdots \end{bmatrix} = -q^{k-1}ac \begin{bmatrix} 0 & \cdots \\ \overline{-k} & \cdots \end{bmatrix} \underbrace{-q^{k-1}(a+c)\begin{bmatrix} 0 & \cdots \\ -k & \cdots \end{bmatrix}}_{\text{smaller terms}}.$$

A cyclic algorithm

Each local rule transports an active column to the next position, plus smaller terms, for some order on two-line configurations.
We solve the system by induction, moving the active column cyclically:

$$C = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & -2 \end{bmatrix}$$

A cyclic algorithm

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We solve the system by induction, moving the active column cyclically:

$$C = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & -2 \end{bmatrix}$$
$$= t \begin{bmatrix} 0 & 2 & 0 \\ 1 & 0 & -2 \end{bmatrix} + \text{smaller terms}$$

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$$C = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & -2 \end{bmatrix}$$

$$= t \begin{bmatrix} 0 & 2 & 0 \\ 1 & 0 & -2 \end{bmatrix} + \text{smaller terms}$$

$$= t \begin{bmatrix} 0 & 0 & 2 \\ 1 & -2 & 0 \end{bmatrix} + \text{smaller terms}$$

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$$= t \begin{bmatrix} 0 & 2 & 0 \\ 1 & 0 & -2 \end{bmatrix} + \text{smaller terms}$$

$$= t \begin{bmatrix} 0 & 0 & 2 \\ 1 & -2 & 0 \end{bmatrix} + \text{smaller terms}$$

$$= t \begin{bmatrix} 0 & 0 & -2 \\ 1 & -2 & 0 \end{bmatrix} + \text{smaller terms}$$

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Each local rule transports an active column to the next position, plus smaller terms, for some order on two-line configurations.
We solve the system by induction, moving the active column cyclically:

$$C = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & -2 \end{bmatrix}$$

$$= t \begin{bmatrix} 0 & 2 & 0 \\ 1 & 0 & -2 \end{bmatrix} + \text{smaller terms}$$

$$= t \begin{bmatrix} 0 & 0 & 2 \\ 1 & -2 & 0 \end{bmatrix} + \text{smaller terms}$$

$$= t \begin{bmatrix} 0 & 0 & -2 \\ 1 & -2 & 0 \end{bmatrix} + \text{smaller terms}$$

$$= t^2 \begin{bmatrix} 0 & -2 & 0 \\ 1 & 0 & -2 \end{bmatrix} + \text{smaller terms}$$

A cyclic algorithm

Each local rule transports an active column to the next position, plus smaller terms, for some order on two-line configurations.
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A cyclic algorithm

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We solve the system by induction, moving the active column cyclically:

$$C = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & -2 \end{bmatrix}$$

$$= t \begin{bmatrix} 0 & 2 & 0 \\ 1 & 0 & -2 \end{bmatrix} + \text{smaller terms}$$

$$= t \begin{bmatrix} 0 & 0 & 2 \\ 1 & -2 & 0 \end{bmatrix} + \text{smaller terms}$$

$$= t \begin{bmatrix} 0 & 0 & -2 \\ 1 & -2 & 0 \end{bmatrix} + \text{smaller terms}$$

$$= t^2 \begin{bmatrix} 0 & -2 & 0 \\ 1 & 0 & -2 \end{bmatrix} + \text{smaller terms}$$

$$= t^2 \begin{bmatrix} -2 & 0 & 0 \\ 0 & 1 & -2 \end{bmatrix} + \text{smaller terms}$$

$$= qt^2 \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & -2 \end{bmatrix} + \text{smaller terms}$$

A cyclic algorithm

Each local rule transports an active column to the next position, plus smaller terms, for some order on two-line configurations.
We solve the system by induction, moving the active column cyclically:

$$C = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & -2 \end{bmatrix}$$

$$= t \begin{bmatrix} 0 & 2 & 0 \\ 1 & 0 & -2 \end{bmatrix} + \text{smaller terms}$$

$$= t \begin{bmatrix} 0 & 0 & 2 \\ 1 & -2 & 0 \end{bmatrix} + \text{smaller terms}$$

$$= t \begin{bmatrix} 0 & 0 & -2 \\ 1 & -2 & 0 \end{bmatrix} + \text{smaller terms}$$

$$= t^2 \begin{bmatrix} 0 & -2 & 0 \\ 1 & 0 & -2 \end{bmatrix} + \text{smaller terms}$$

$$= t^2 \begin{bmatrix} -2 & 0 & 0 \\ 0 & 1 & -2 \end{bmatrix} + \text{smaller terms}$$

$$= qt^2 \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & -2 \end{bmatrix} + \text{smaller terms}$$

$$(1 - qt^2)\, C = \underbrace{(1-t) \begin{bmatrix} 2 & 0 & 0 \\ 1 & 0 & -2 \end{bmatrix} + t(1-t) \begin{bmatrix} 0 & 0 & -2 \\ 1 & 0 & -2 \end{bmatrix}}_{\text{smaller terms}}.$$

Application 1: nonsymmetric Askey–Wilson polynomials

Theorems A and B give explicit recurrences in the one-variable case ($n = 1$):

Theorem

For $k \geq 1$,

$$F_{(k+1)} = \big(z + u_k\big)F_{(k)} + \big(v_k z + w_k\big)F_{(-k)},$$
$$F_{(-k-1)} = \big(\widetilde{v}_k z^{-1} + \widetilde{w}_k\big)F_{(k)} + \big(z^{-1} + \widetilde{u}_k\big)F_{(-k)},$$

with $u_k = w_k\widetilde{v}_k$, $\widetilde{u}_k = v_k\widetilde{w}_k$, *and*

$$v_k = q^{k-1}\frac{ac + q - acq^k - abcdq^k}{1 - abcdq^{2k-1}}, \qquad w_k = q^k\frac{-(a+c) + ac(b+d)q^k}{1 - abcdq^{2k}},$$
$$\widetilde{v}_k = \frac{1 + bd - bdq^k - abcdq^{k-1}}{1 - abcdq^{2k-1}}, \qquad \widetilde{w}_k = \frac{-(b+d) + bd(a+c)q^k}{1 - abcdq^{2k}}.$$

This result refines the three-term relation for symmetric Askey–Wilson polynomials $P_k = F_{(k)} + F_{(-k)}$.

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This result refines the three-term relation for symmetric Askey–Wilson polynomials $P_k = F_{(k)} + F_{(-k)}$.

All these coefficients are rational functions of q^k. In particular, for $n = 1$, all two-line configurations are given by 8 rational functions of q^k. For any n, there are only finitely many rational functions to compute.

Application 2: top homogeneous parts and type A

Definition

A two-line configuration is simple if all its labels are regular (no bars), and two nonzero labels in the same column have the same sign.

Example

$$\begin{bmatrix} 2 & -2 & 0 & 0 & 3 & 0 \\ 3 & -2 & 1 & -1 & 0 & 2 \end{bmatrix} \text{ is simple,}$$

$$\begin{bmatrix} 0 & -3 & 2 & 0 & 0 & -2 \\ 2 & -1 & 0 & \overline{3} & -2 & \overline{1} \end{bmatrix} \text{ is not.}$$

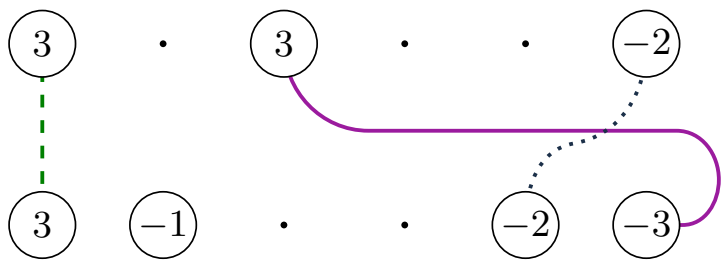
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Proposition

A simple two-line configuration $\begin{bmatrix} \sigma \\ \tau \end{bmatrix}$ *is the generating function of reflected two-line queues with top row* σ *and bottom row* τ*.*

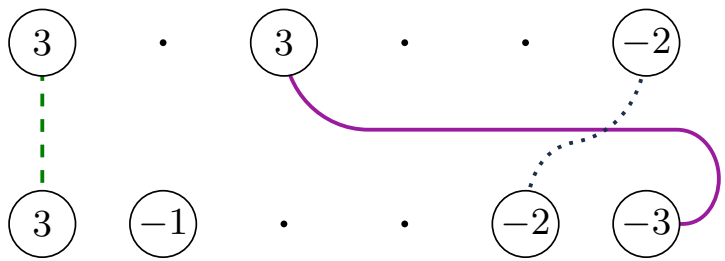


$$\begin{bmatrix} 3 & 0 & 3 & 0 & 0 & -2 \\ 3 & -1 & 0 & 0 & -2 & -3 \end{bmatrix}$$

Application 2: top homogeneous parts and type A

Proposition

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Together with the two-row recurrence (Theorem A), this gives:

Theorem C (top homogeneous parts)

For $\mu \in \mathbb{Z}^n_{\geq 0}$,

$$\mathrm{Top}(F_\mu) = F^A_\mu, \qquad \mathrm{Top}\left(E_\mu(z_1,\ldots,z_n;q,t)\right) = E^A_{\mu^{\mathrm{rev}}}(z_n,\ldots,z_1;q^{-1},t^{-1}).$$

Main results (BD–Williams)

Theorem A (two-row recurrence)

For $\kappa \in \mathbb{Z}^n$, *set* $(\kappa^-)_i = \kappa_i - \mathrm{Sign}(\kappa_i)$. *Then*

$$F_\kappa = \sum_{(\nu,\mu)} a_\kappa^\mu(\nu)\, z^\nu F_\mu,$$

where $\mu \in W_0 \cdot \kappa^-$ *and* $\nu \in \mathbb{Z}^n$ *with* $\nu_i \in \{0, \mathrm{Sign}(\kappa_i)\}$.

Theorem B (local rules)

The coefficients $a_\kappa^\mu(\nu)$ *are uniquely characterized by a family of local rules.*

Theorem C (top homogeneous parts)

For $\mu \in \mathbb{Z}^n_{\geq 0}$,

$$\mathrm{Top}(F_\mu) = F_\mu^A, \qquad \mathrm{Top}\left(E_\mu(z_1,\ldots,z_n;q,t)\right) = E^A_{\mu^{\mathrm{rev}}}(z_n,\ldots,z_1;q^{-1},t^{-1}).$$